Table 6. Robustness and falsification summary

Test	Result	Implication
Leave-one-ETF-out	External-financing activation remains negative i	Not driven by one fund
Architecture assignment permutation	External finance two-sided p = 0.0148; growth-d	Cross-sectional alignment is non-random
Alternative definitions	External finance expected-sign rate = 0.867; co	Sign stability across measurement choices
Sorted portfolios	External finance HML = -0.01017 (p = 0.0198);	Portfolio-level economic relevance
Factor-beta controls	Signs persist but precision weakens for external	Part of activation overlaps with conventional styles
2020Q4 boundary validation	Primary-channel Spearman ρ = 0.59–0.86; conc	Primary channels remain ordered; extended comp
2020–2025 aligned panel	Growth-duration = −0.0548 at six months and −	Principal dynamic magnitude persists near the ho 01
Future-CPU lead placebo	No lead survives BH adjustment	No support for systematic pre-trend; causal claims
Downside and quantile models	Supporting signs; bootstrap intervals often wide	Secondary, not anchor evidence

Note. HML denotes the high-minus-low architecture portfolio. Multiple-testing adjustments are applied within the pre-specified test families.